

Technical Report: Power BI Dashboard Enhancement

Data Science CA1 - Task 3: Financial Analysis

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May 7, 2026

1 1. The Audit (Initial Assessment)

The original dashboard (`codal_news_dashboard.pbix`) was designed to examine the impact of news on trading behavior. Upon detailed inspection, the following weaknesses and limitations were identified:

- **Incomplete Data Model:** Only the `khodro_final_daily` table existed, without a separate `Date` table. This made it impossible to use Time Intelligence functions, such as calculating dynamic moving averages based on user-selected intervals.
- **Lack of DAX-Calculated Technical Indicators:** All technical analyses (like RSI and Bollinger band hits) relied on pre-calculated columns in the CSV file. The dashboard could not dynamically calculate indicators like 20/50-day SMAs or Bollinger Bands based on historical ranges selected by the user.
- **Absence of Primary Price and Volume Charts:** The most basic tools of technical analysis—a line chart for the closing price (`close`) and a bar chart for trading volume (`vol`)—were missing.
- **Limited Time Filtering:** There was only one Slicer for the `days_to_news` field, with no Slicer provided for selecting a custom date range (From/To). Users could not, for instance, analyze just the last 6 months.
- **Static and Single-Purpose Design:** The two existing pages were built solely around the "impact of days prior to news," failing to cover general price trends, volatility, or buy/sell signals.
- **Poor User Experience (UX):** There were no interactive Tooltips, Bookmarks for navigating views, KPI cards (key performance indicators), or standard stock market color coding (red for decrease, green for increase).

2 2. The Solution

To resolve the aforementioned weaknesses, the following modifications were applied to the `codal_news_dashboard.pbix` file:

2.1 2.1. Data Model Improvement: Creating a Calendar Table

Using DAX, a dynamic date table named `DateTable` was created to cover the minimum to maximum dates present in the dataset. A one-to-many (*:1) relationship was then established between the `date` column in the main table and the `Date` column in the calendar table. This forms the backbone of all time-driven analyses.

2.2 2.2. Implementing Technical Indicators with DAX

Three main groups of indicators were implemented using DAX to replace the dependency on static CSV values:

A) Simple Moving Average (SMA 20 & 50)

The 50-day moving average was also created using the same structure and a -50 window.

```
1 SMA 20 = CALCULATE(  
2     AVERAGE('khodro_final_daily'[close]),  
3     DATESINPERIOD('DateTable'[Date], LASTDATE('DateTable'[Date]),  
4         -20, DAY)  
5 )
```

B) Bollinger Bands

First, the 20-day standard deviation was calculated, followed by the creation of the upper and lower bands at a distance of twice the standard deviation from the SMA 20:

```
1 StdDev 20 = CALCULATE(  
2     STDEV.P('khodro_final_daily'[close]),  
3     DATESINPERIOD('DateTable'[Date], LASTDATE('DateTable'[Date]),  
4         -20, DAY)  
5 )  
6 Upper Band = [SMA 20] + 2 * [StdDev 20]  
7 Lower Band = [SMA 20] - 2 * [StdDev 20]
```

C) Relative Strength Index (RSI 14)

To calculate the RSI, daily changes (Price Change) and subsequent gains and losses were defined. The 14-day average of gains and losses was obtained using **AVERAGEX** over a 14-day window, and finally, the RSI was calculated using the standard formula (using simple moving average as an acceptable approximation):

```
1 Price Change = SUM('khodro_final_daily'[close]) - CALCULATE(SUM('khodro_final_daily'[close]), PREVIOUSDAY('DateTable'[Date]))  
2 Gain = IF([Price Change] > 0, [Price Change], 0)  
3 Loss = IF([Price Change] < 0, -[Price Change], 0)  
4 AvgGain14 = AVERAGEX(DATESINPERIOD('DateTable'[Date], LASTDATE('DateTable'[Date]), -14, DAY), [Gain])  
5 AvgLoss14 = AVERAGEX(DATESINPERIOD('DateTable'[Date], LASTDATE('DateTable'[Date]), -14, DAY), [Loss])  
6 RSI14 = VAR RS = IF([AvgLoss14] = 0, 100, DIVIDE([AvgGain14], [AvgLoss14]))  
7 RETURN 100 - 100 / (1 + RS)
```

2.3 2.3. Complete UI Redesign

A new page named **Main Dashboard** was created, featuring a professional financial layout:

- **4 KPI Cards (Top):** Last Close (custom Measure), SMA 20 (latest value), RSI14 (latest value), and Daily Change % (calculated by intelligently locating the last trading day).
- **Main Line Chart:** Displays the closing price alongside SMA 20, SMA 50, and Bollinger Bands (Upper/Lower). Each line uses distinct colors and patterns (e.g., dashed lines for SMA 50 and bands).
- **RSI14 Chart:** A separate line chart positioned below the main chart with reference lines at the 30 and 70 levels (green and red) to highlight overbought/oversold signals.
- **Volume Chart:** A column chart visually displaying trading volume alongside other charts.
- **Dynamic Date Slicer:** A 'Between' date slicer allows users to select custom date ranges. All Measures and charts recalculate automatically based on this selection.
- **Interactive Tooltip:** A dedicated Tooltip page (TooltipPage) was built. Hovering over any point on the price chart displays daily details (Date, Price, SMA20, RSI14, Volume) in a compact window.
- **Theme & Aesthetics:** A dark background (#1E1E1E), rounded borders, white titles, and finance-appropriate colors (blue for averages, light red for bands, green/red for RSI) were chosen to enhance readability and intuitive data comprehension.

3 3. Insight Discovery

With the enhanced dashboard, advanced analytical questions that the initial dashboard could not handle can now be answered:

New Analytical Question	Response Mechanism in Enhanced Dashboard
Is the trend bullish or bearish?	The crossover of the 20 and 50 SMAs (Golden/Death Cross) is directly visible on the chart.
Is the price on the verge of a volatility breakout?	Assessed by observing the price exiting (and returning to) the Bollinger Bands, or frequently approaching a specific band.
Is the stock in an overbought or oversold state?	Shown on the dedicated RSI14 chart. Crossing above 70 warns of a sell, while dropping below 30 signals a buy.
What was the strength of buyers and sellers over the past 14 days?	The numeric RSI14 value in the KPI card and its position between 0 and 100 indicate relative strength.
How did the price react on days with extremely high trading volume?	The volume column chart alongside the price chart enables the identification of high-volume days and their impact on price trends.
How did the indicators trend over a custom time frame (e.g., 3 months, 1 year)?	The 'Between' Date Slicer filters all charts and KPI cards based on the user's selected range, providing cross-sectional analysis.
Where is the key data for each trading day (Price, SMA, RSI) without cluttering the chart?	The custom Tooltip page instantly displays a summary of that day's information upon mouse hover.

4 4. Conclusion

Despite containing rich data, the initial dashboard lacked fundamental technical analysis tools and a proper user experience. By adding a complete data model (Calendar Table), implementing moving averages, Bollinger Bands, and RSI via DAX, and executing a visual redesign based on UX/UI principles, the dashboard was transformed into a dynamic and professional stock analysis tool. Analysts can now easily evaluate market trends, volatility, and momentum across any timeframe to make more informed decisions.